

# Homework Assignment 10

Connor Mooneyhan

November 14, 2025

**4B.2.** Suppose  $f \in \mathcal{L}^1(\mathbb{R})$ . Prove that

$$\limsup_{t \downarrow 0} \left\{ \frac{1}{|I|} \int_I |f - f_I| : I \text{ is an interval of length } t \text{ containing } b \right\} = 0$$

for almost every  $b \in \mathbb{R}$ .

*Proof.* Let  $t > 0$ . Then

$$\int_I |f - f_I| \leq \int_I |f - f(b)| + \int_I |f(b) - f_I|. \quad (1)$$

Note that  $I \subset [b - t, b + t]$ , so we can say

$$\begin{aligned} \frac{1}{t} \int_I |f - f(b)| &\leq \frac{1}{t} \int_{b-t}^{b+t} |f - f(b)| \\ &= 2 \left( \frac{1}{2t} \int_{b-t}^{b+t} |f - f(b)| \right), \end{aligned}$$

which goes to 0 as  $t$  goes to 0 from the right. Also,

$$\begin{aligned} \frac{1}{t} \int_I |f(b) - f_I| &= |f(b) - f_I| \frac{1}{t} \int_I 1 \\ &= \left| \frac{1}{t} \int_I f(b) - \frac{1}{t} \int_I f \right| \\ &\leq \frac{1}{t} \int_I |f - f(b)| \\ &\leq 2 \left( \frac{1}{2t} \int_{b-t}^{b+t} |f - f(b)| \right). \end{aligned}$$

Given  $\varepsilon > 0$ , there is a  $\delta > 0$  such that if  $0 < t < \delta$  then

$$\frac{1}{2t} \int_{b-t}^{b+t} |f - f(b)| < \frac{\varepsilon}{4}$$

by the Lebesgue Differentiation Theorem. Thus

$$\sup \left\{ \frac{1}{|I|} \int_I |f - f_I| : I \text{ is an interval of length } t \text{ containing } b \right\} < \varepsilon$$

for  $0 < t < \delta$ . Since this is true for any such  $\varepsilon$ , the proof is complete.  $\square$

**4B.1.** Suppose  $f \in \mathcal{L}^1(\mathbb{R})$ . Prove that

$$\lim_{t \downarrow 0} \frac{1}{2t} \int_{b-t}^{b+t} |f - f_{[b-t, b+t]}| = 0$$

for almost every  $b \in \mathbb{R}$ .

*Proof.* We have

$$\begin{aligned} & \lim_{t \downarrow 0} \frac{1}{2t} \int_{b-t}^{b+t} |f - f_{[b-t, b+t]}| \\ & \leq \lim_{t \downarrow 0} \sup \left\{ \frac{1}{|I|} \int_I |f - f_I| : I \text{ is an interval of length } 2t \text{ containing } b \right\} \\ & = 0 \end{aligned}$$

by 4B.2. □

**4B.4.** Prove that the Lebesgue Differentiation Theorem still holds if the hypothesis that  $\int_{-\infty}^{\infty} |f| < \infty$  is weakened to the requirement that  $\int_{-\infty}^x |f| < \infty$  for all  $x \in \mathbb{R}$ .

**7A.1.** Suppose  $\mu$  is a measure. Prove that

$$\|f + g\|_{\infty} \leq \|f\|_{\infty} + \|g\|_{\infty} \quad \text{and} \quad \|\alpha f\|_{\infty} = |\alpha| \|f\|_{\infty}$$

for all  $f, g \in \mathcal{L}^{\infty}(\mu)$  and all  $\alpha \in \mathbb{F}$ . Conclude that with the usual operations of addition and scalar multiplication of functions,  $\mathcal{L}^{\infty}(\mu)$  is a vector space.

*Proof.* Given  $x \in X$ , we have

$$\begin{aligned} |f(x) + g(x)| & \leq |f(x)| + |g(x)| \\ & \leq \|f\|_{\infty} + \|g\|_{\infty} \text{ almost everywhere.} \end{aligned}$$

It follows that  $\|f + g\|_{\infty} \leq \|f\|_{\infty} + \|g\|_{\infty}$  almost everywhere. □

**7A.2.** Suppose  $a \geq 0$ ,  $b \geq 0$ , and  $1 < p < \infty$ . Prove that

$$ab = \frac{a^p}{p} + \frac{b^{p'}}{p'}$$

if and only if  $a^p = b^{p'}$ .

*Proof.* As in the proof of Young's Inequality, we fix  $b > 0$  and define  $f : (0, \infty) \rightarrow \mathbb{R}$  by

$$f(a) = \frac{a^p}{p} + \frac{b^{p'}}{p'} - ab.$$

As noted in that proof, the global minimum is at  $b^{1/(p-1)}$ , so setting  $p' = p/(p-1)$ , we get  $f(b^{1/(p-1)}) = 0$ . The desired result is then obtained by writing this as  $a = b^{1/(p-1)}$ , and then raising to the  $p^{\text{th}}$  power to get

$$a^p = b^{\frac{p}{p-1}} = b^{p'}.$$

□

**7A.4.** Prove Hölder's inequality in the cases  $p = 1$  and  $p = \infty$ .

*Proof.* We have

$$\begin{aligned}\|fg\|_1 &= \int_X |fg| \\ &= \int_X |f||g| \\ &\leq \int_X |f| \|g\|_\infty \text{ almost everywhere} \\ &= \|g\|_\infty \int_X |f| \\ &= \|g\|_\infty \|f\|_1 \\ &= \|f\|_1 \|g\|_\infty,\end{aligned}$$

which proves both cases. □